

Azi Khan

CS601

Quiz 2

For More ph# 03217811145



Remember in prayers

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Cs601

Quiz no 2

1. In _____ each station is allocated a time slot during which it can send data

TDMA.....confirm

2. PAP and CHAP are used by _____ as Authentication Protocols.

PPP.....confirm

3. In each character protocol each character of frame is encoded using _____ coding system .

Bit-oriented

4. IN MAC, when nodes use a multi-point or broadcast link, we need a _____ to coordinate access to the link.

Multiple-access protocol/.....confirm

5. The original Ethernet technology with the data rate of 10 MBps is called _____?

Standard Ethernet....confirm

6. _____ uses fiber-optic cable

10base-F....confirm

7. One of the most common protocols for point-to-point access is _____ .

PPP.....confirm

8. is an Authentication Protocol, which uses two-step process to authenticate user information.

PAP.....confirm

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9. If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause ____ number of bits being impacted by the noise

More confirm

10. For carrier sense Multiple Access/Collision Detection (CSMA/CD), we need a restriction on the

Collision size....confirm

11. Stop-and-Wait protocol is used for _____.

Both flow and error control.....confirm

12. In hexadecimal notation of Ethernet address, each byte is separated by ____ symbol.

Colon....confirm

13. In a linear block code, the ____ of any two valid codewords creates another valid codeword.

14.

XOR.....confirm

15. In the ____ method, the stations in a network are organized in a logical ring.

Token-passing....confirm

16. Which one of the following is not a Channelization Protocol?

CSMA.....confirm

17. In order to find the location of two errors in an eight bit stream, we have to see ____ different combinations.

28....confirm

18. If we need to correct a single error in an 8-bit data unit, we need to consider _____ possible error locations

16confirm

19. In _____ each station sends a frame whenever it has a frame to send.

Slotted ALoHA confirm

20 _____ is the process of converting digital data to digital signals.

Line coding confirm

21. In case of higher data rate, the number of impacted bits will be _____.

Higher

22. In _____ protocol, the sender sends one frame at a time and wants for an acknowledgment before sending the next one.

Stop-and-wait

23. In block coding we divide our message into blocks of K bits called as _____.

Data wordsconfirm

24. CDMA stands for _____.

Code division Multiple Accessconfirm

25 _____ is not a function of data link control.

Modulationconfirm

26. CHAP is an Authentication Protocol, which uses _____ process to authenticate user information.

Three-way.....confirm from net

27. In the context of Ethernet address, the source address is always a _____ address.

Unicast...confirm

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28. In the context of Ethernet address, the destination address is always a _____ address.

Unicast, multicast, broadcast....confirm

29 _____ transmission technology (based on Ethernet) provides a data rate of 1 billion bits per second.

Gigabit Ethernet....confirm

30. Fast Ethernet has a data rate of _____ Mbps.

100.....confirm

31. In random access method, when two or more than two nodes transmit at the same time when

All the nodes in collision survives

32. Data link layer is divided into _____ sub layers

Two.....confirm

33. If the hamming distance between sent and received code word is _____ then it shows that received data is corrupted.

(Is not equal to)0....confirm

34. In order to find hamming distance between two pair of words, _____ logical operation is used.

XOR....confirm

35. Ethernet address is of _____ bytes

6....confirm

36. In ____ a station monitors the medium after it sends a frame to see if the transmission was successful.

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CSMA/CD.....confirm

37. For carrier sense Multiple Access/Collision Detection (CSMA/CD), we need a restriction on the ____.

Frame size...confirm

38. Ethernet address is of ____ bytes

6

39. What was the main drawback in the aloha random access method

Redundancy in data

40. In telephone, network POP stands for

Point-of presence

41. Possession of token gives the station the right to ____.

Send the data

42. In Ethernet, the source address field in the MAC frame is the _ address

Previous station physical

43. Normally the value of Kmax in pure aloha is

15

44 ____ uses thin coaxial cable.

10 base 5

45. MAC stands for ____

Media access control

46. In the context of Ethernet implementations maximum medium length of 10 base-F

2000

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47. Transmission technology based on Ethernet provides a data rate of 1 billion bits per second.

Standard Ethernet

48. Works with topologies in which one device is designated as a primary station and the other devices are secondary stations

None of the given

49. MAC address is of _____.

48 bits

50. The type of aloha which improves the efficiency of pure aloha is.

Slotted aloha

51. A personal computer or workstation on an Ethernet network must have _____ card

NIC

52. In which controlled access method all data exchange must be made through primary device even when the ultimate destination is a secondary device

Polling

53. In _____ each band is reserved for a specific station and it belongs to the station all the time

FDMA

54. In pure aloha the vulnerable time is _____ the frame transmission time

Two times

55. DSL stands for _____.

Digital subscriber line

56. In the context of Ethernet address, the source address is always a ____ ____ address

Multicast

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57. The original Ethernet technology with the data rate of 10 Mbps is called

Standard ethernet

58. Which of the following is a correct controlled access method

All options

59. When we represent the data in CDMA if a station is idle then it sends

0

60. In telephone, network POP stands for ____.

Point of presence

61. What was the main drawback in the aloha random access method

Redundancy data rate

62. _transmission technology based on Ethernet provide a data rate of 1 billion bits per second

Gigabit Ethernet

63. In which controlled access method all data exchanges must be made through a primary device even when the ultimate destination is a secondary device.

Polling

64. In telephone, network POP stands for

Point of presence

65. Mac address is of ____

48 bits

66. In ____ collision are avoided by deferring transmission even if the channel is idle

The interframe space

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67. POTS stand for _?

Plain old telephone system.....confirm

68. Normally the value of Kmax in pure aloha is

15

69. In Ethernet frame both destination and sender addresses are of length

6 byte

70. In _____ each band is reserved for a specific station and it belongs to the station all the time

FDMA

71. The original Ethernet technology with a data rate of 10MBps is called

Standard Ethernet

72. The _ method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately

1 persistent method

73. IN allocated a time slot during which it can send data each station is

TDMA

74. In which controlled access method all data exchanges must be made through primary device even when the ultimate destination is a secondary device

Polling

75. CDMA stands for

Code data multiply access

76. Works with topologies in which one device is designated as a primary station and the other device are secondary stations

Polling

77. In Ethernet, the source address field in the MAC frame is the address

Next destination physical

1. VCI address is _____ address in virtual circuit approach.

1. Private
2. Variable
3. Local
4. Global

2. Traditionally MANs have been implemented using one of the 2 technologies, Circuit Switching and Packet Switching

1. true
2. false

3. In the _____ method, the stations in a network are organized in a logical ring.

1. Token-passing
2. Polling
3. Reservation
4. None of the given

4. Like 10 Base 5, 10 Base 2 is a _____ topology LAN.

1. Ring
2. Mesh
3. Star
4. **Bus**

5. In Pure ALOHA, the vulnerable time is _____ the frame transmission time.

1. Same as
2. Three times
3. **Two times**
4. Four times

6. The transmission medium that carries the message is referred to as the _____.

1. send and receive device
2. **communication channel**
3. protocol
4. gateways

7. Analog signals are refers to be _____

1. **Continuous**
2. Discrete
3. Periodic

4. Aperiodic

8. There are _____ types of serial transmission.

1. 3
2. 4
3. 2
4. None of the given

9. To calculate the data rate for noisy channel _____ formula is used.

1. Shannon
2. Nyquist
3. Propagation
4. Greedy

10. There are three types of Hubs that exist. Which of the following options correctly describes these three?

1. Passive, dormant, special
2. Active , dormant , passive
3. Passive, Active, Turbo
4. Passive, Active, Intelligent

11. _____ is the rate of change with respect to time.

1. Voltage

- 2. Time
- 3. Frequency
- 4. Amplitude**

12. To control and manage the transfer of data, a protocol called Point to Point Protocol (PPP) is used at the_____layer.

- 1. Physical
- 2. Network**
- 3. Presentation
- 4. Data link

13. What is the frequency of a sine wave if it completes one cycle in 2 seconds?

- 1. 0.5**
- 2. 0.6
- 3. 0.3
- 4. 0.2

14. _____relatively measures the strength of two signals.

- 1. Signal rate
- 2. Bit rate
- 3. Decibel
- 4. Pulse Rate**

15. Central Hub in a Tree is an_____.

1. Active hub

2. passive hub

16. At the CRC generator, _____ added to the data unit before the divisional process.

1. 0s are

2. 1s are

3. a polynomial is

4. a CRC remainder is

17. A broadcast signal is received by the orbiting satellites which rebroadcasts _____.

1. to the resender

2. to the sender

3. to the atmosphere

4. none of the given

18. The flag in PPP is a byte that needs to be _____ whenever it appears in the data section of the frame.

1. Duplicated

2. Blocked

3. Cleaned

4. Escaped

19. In _____, a signal is directed straight from antenna to antenna.

1. Line of sight propagation

2. Ground propagation

3. Space propagation

4. Tropospheric propagation

20. Which of the following consist of just one redundant bit per data unit?

1. Two dimensional Parity check.

2. CRC

3. Simple Parity Check

4. Checksum

21. The term _____ means that only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.

1. Packet Error

2. Burst Error

3. Single Bit Error

4. Character Error

22. Flow control is needed to prevent_____.

1. Bit errors

2. Overflow of the sender buffer

3. Overflow of the receiver buffer

4. Collision between sender and receiver

23. **Transmission Control Protocol is a(n)_____ layer protocol.**

1. Transport
2. Application
3. Session
- 4. Network**

24. **A fiber -optic cable transmit signals in the form of_____.**

- 1. Light**
2. Sound
3. Wave
4. None of the given

25. **Possession of TOKEN gives the station the right to_____.**

1. Modify the data
- 2. Send the data**
3. Delete the data
4. Receive the data

26. **The inversion of the level at 1 bit is called as_____.**

1. NRZ-L
- 2. NRZ-I**
3. RZ

4. None of the given.
27. **What was the main draw back in ALOHA Random-Access method?**
1. Redundancy in Data
 2. Potential Collision
 3. Inconsistent Data rate
 4. All of the given
28. **Data is transmitted using light through a _____ cable.**
1. twisted pair
 2. fiber-optic
 3. coaxial
 4. microwave
29. **In _____, each station sends a frame whenever it has a frame to send.**
1. Slotted ALOHA
 2. Fragmented ALOHA
 3. Pure ALOHA
 4. None of the given
30. **In OSI model headers are added at layer _____.**
1. 1,2,3 only
 2. 6,5,7 only
 3. 5,4,3 only

4. 6,5,4,3,2 only

31. Is there any difference between Data communication and telecommunication networks?

- 1. Yes
- 2. No

32. ____ is an Authentication Protocol, which is used by the PPP to authenticate passwords.

- 1. CHAP
- 2. PAP
- 3. LCP
- 4. ICMP

33. Which level of the TCP/IP reference model routes data/information across a network channel?

- 1. Application Layer
- 2. Data Link Layer
- 3. Transport Layer
- 4. Network Layer

34. In ASK, both _____ and _____ remain constant.

- 1. Amplitude, Frequency
- 2. Amplitude, phase
- 3. Frequency, Phase

4. Amplitude, Samples

35. Which error detection method consists of a parity bit for each unit as well as an entire data unit of parity bits?

- 1. Checksum
- 2. Cyclic Redundancy Check (CRC)
- 3. Longitudinal Red Check (LRC)**
- 4. Vertical Redundancy Check (VRC)

36. Baud rate is greater than or equal to the bit rate.

- 1. True
- 2. False**

37. The most common type of connector used by coaxial cable is _____.

- 1. BNC**
- 2. RJ-45
- 3. RJ-11
- 4. RJ-57

38. In even parity check code, the value of syndrome is _____ if the number of 1s is even.

- 1. 1**
- 2. 2
- 3. 0**
- 4. 4

39. What is the period of a Sine wave having frequency of 10 Hz?

1. 0.2

2. 0.1

3. 0.5

4. 0.3

40. In __, both peak amplitude and frequency remains constant as the phase changes.

1. PSK

2. QAM

3. Both PSK & QAM

4. None of the given

41. There are _____ types of guided media.

1. 2

2. 3

3. 4

4. 5

42. _____ is responsible for governing node to node communication.

1. Application Layer

2. Session Layer

3. Data Link layer

4. Presentation Layer

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43. TROPOSHERE is the layer of atmosphere above the troposphere but below space

1. True

2. False

44. Terminal, microprocessor, computer, printer or any other device that generates or consumes digital data is known as _____

1. DCE

2. DTE

3. DCE & DTE

4. None of the given

45. Normally the value of K_{max} in pure ALOHA is _____.

1. 25

2. 10

3. 15

4. 30

46. Sampling means measuring _____ of signal at equal intervals

1. Amplitudes

2. Frequencies

3. Phases

4. None of the given

47. A sine wave is _____.

1. periodic and discrete
2. aperiodic and discrete
- 3. periodic and continuous**
4. aperiodic and continuous

48. The inversion of the level represents a 1 bit in _____ encoding

- 1. NRZ-I**
2. NRZ-L
3. RZ
4. Manchester

49. To make sure that source does not overwhelm destination by sending data faster than it can be handled and processed is called?

1. Addressing & Routing
2. Exchange Management
- 3. Flow Control**
4. Recovery

50. Time domain plot show changes in signal phase with respect to time.

1. true
- 2. false**

51. In which controlled-access method all data exchanges must be made through primary device even when the ultimate destination is a secondary device?

1. Polling
2. Reservation
3. Token Passing
4. Port Forwarding

52. If a digital signal has "L" number of levels, _____ is the number of bits required to represent each level.

1. Log L
2. Log Base 2 L
3. L Square
4. Log(2*L)

53. _____ category of coaxial cable is used for Cable TV.

1. RG-58
2. RG-59
3. RG-11
4. RG-47

54. The level of signal is always positive in NRZ encoding.

1. true
2. false

55. Radio is an example of _____ signal conversion.

1. Analog to analog
2. Analog to digital

3. Digital to Digital

4. Digital to analog

56. **Analog To Analog Conversion Methods include_____.**

1. AM, PM and FM

2. AM, FSK and PM

3. AM, PM and QAM

4. None of the given

57. **The original Ethernet technology with the data rate of 10 Mbps is called_____?**

1. Standard Ethernet

2. Fast Ethernet

3. Gigabit Ethernet

4. 10 Gigabit Ethernet

58. **Which of the following is not a characteristic of a sine wave?**

1. Amplitude

2. **Segmentation**

3. Phase

4. Frequency

59. **By using_____system, change in the file contents during file transmission can be avoided.**

1. Routing

2. Flow Control

3. Error Detection and Correction

4. Congestion Control

60. In order to find the location of two errors in an eight bit stream, we have to see _____ different combinations.

1. 28

2. 16

3. 8

4. 4

61. In CSMA/CA, _____ is the amount of time divided into slots.

1. Interframe Window

2. Contention Window

3. Fragmented Window

4. Collided Window

62. ____ is used in the Stop-and-Wait protocol to overcome the issue of duplication.

1. Frame sequencing

2. Acknowledgement numbering

3. Counter and Timer reset

4. Frame sequencing and Acknowledgement numbering

63. Which one of the following is a correct-access method?

1. Polling
2. Token Passing
- 3. Reservation**
4. All of the given

64. In the encoding where positive to negative transition is one and vice versa is zero.

1. Manchester
2. Differential Manchester
3. All of the given
- 4. None of given**

65. If the window size is 63 in go-back N ARQ then what is the range of sequence number?

- 1. 0 to 63**
2. 0 to 64
3. 1 to 63
4. 1 to 64

66. Following characteristics fully describe _____,

Amplitude

Period/Frequency

Phase

- 1. Sine Waves**

2. Digital Signals
 3. Aperiodic Signals
 4. None of the given
67. **Digitizing an Analog Signal is called_____.**
1. **Analog-to-Digital Conversion**
 2. Digital-to-Digital Conversion
 3. Digital-to-Analog Conversion
 4. None of the given
68. **At the CRC checker,_____means that the data unit is damaged.**
1. A string of 0s
 2. A string of 1s
 3. A string of alternating 1s and 0s
 4. **A nonzero remainder**
69. **Which one of the following is a Random-Access protocol?**
1. TDMA
 2. **CDMA/CD**
 3. ALOHA
 4. CDMA/CA
70. **Human voice is example of_____signal.**
1. Digital

2. Analog

3. Discrete

4. None of the given

71. In statistical time division multiplexing, number of slots in frame are less than _____.

1. Output data rate

2. Input lines

3. Input frequency

4. Multiplexed T lines

72. A _____ modem is used to connect two DTEs directly.

1. 'NULL

2. Cable

3. DSL

4. None of the given

73. If we need to correct a single error in an 8-bit data unit, we need to consider _____ possible error locations.

1. 16

2. 2

3. 4

4. 8

74. Asynchronous TDM is efficient only when the size of the time slot is kept relatively_____.

- 1. Large**
2. Small
3. Medium
4. Zero

75. The_____protocol uses both flow and error control.

1. TCP/IP
2. HDLC
3. ICMP
- 4. Stop-and-Wait**

76. In_, each station is allocated a time slot during which it can send data.

1. CDMA
2. NDMA
- 3. TDMA**
4. FDMA

77. The extra bits added with the original data for error detection/correction are called_____.

1. Fault bits
2. Port numbers

3. Redundant bits

4. Process IDs

78. ____ is correct formula to determine the total number of ports needed for one system connected in a mesh network.

1. $N+1$

2. $N-1$

3. $N(N+1)$

4. $N \times N$

79. _____ is sometimes called the bit rate.

1. signal rate

2. modulation rate

3. **Data rate**

4. pulse rate

80. Collisions in CSMA/CA are avoided through _____.

1. The Interframe Space

2. The Contention Window

3. Acknowledgements

4. **All of the given**

81. _____ category of coaxial cable is used for thin Ethernet.

1. **RG-58**

2. RG-59

3. RG-1

4. RG-47

82. Which one is not the function of data link layer?

1. Line discipline

2. Flow control

3. Error control

4. Network control

83. Session layer is responsible for _____.

1. Reassembly of data

2. Maintaining the connection

84. _____ mode of serial transmission guarantees fixed rate of data.

1. Synchronous

2. Asynchronous

3. Isochronous

4. Metasynchronous

85. Data Link layer provides services to the _____ layer.

1. Application

2. Transport

3. Network

4. Session

86. The _____ is the number of signal elements sent per unit time.

1. Data rate

2. Signal rate

3. Bit rate

4. Period

87. The logical connection between the peer layers is _____ connection.

1. Physical

2. Direct

3. Indirect

4. Tangible

88. What is the period of a Sine wave having frequency of 5 Hz?

1. 0.2

2. 0.1

3. 0.5

4. 0.3

89. A sine wave is defined by _____ characteristics.

1. 2

2. 3

3. 4

4. 5

90. If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause _____ number of bits being impacted by the noise.

1. Constant
2. More
3. Less
4. Zero

91. S-frames in High-Level Data Link Control (HDLC) are only used to transfer _____ information.

1. User data
2. Control
3. Redundant
4. Original

92. The _____ generates the data and passes it along with any control information to a _____.

1. DTE, DCE
2. DCE, DTE
3. DTE, DTE
4. DCE, DCE

93. The type of ALOHA which improves the efficiency of Pure ALOHA is _____.

1. Slotted ALOHA

2. Upper ALOHA

3. Integrated ALOHA

4. Lower ALOHA

94. What is the period of a Sine wave having frequency of 2 Hz?

1. 0.2

2. 0.1

3. 0.5

4. 0.3

95. WDM stands for _____.

1. Wideband De-Modulation

2. Worst Data Manipulation

3. Wavelength Division Multiplexing

4. None of the given

96. Latency is made of _____ components.

1. Two

2. Three

3. Four

4. Five

97. The logical connection between the peer layers is _____ connection.

1. Physical

- 2. Direct
- 3. Indirect
- 4. Tengible**

98. In line discipline after the data transmission, the sending system finishes with an_____frame.

- 1. EOT**
- 2. EKT
- 3. ENT
- 4. ESP

99. ____ of a signal is the collection of all the component frequencies it contains.

- 1. Time domain
- 2. Frequency domain
- 3. Frequency spectrum**
- 4. None of the given

100. CHAP is an Authentication Protocol, which uses_____process to authenticate user information.

- 1. Two-way
- 2. Three-way**
- 3. Four-way
- 4. Five-way

101. RZ stands for_____.

- 1. Return to Z
- 2. Retire a Zero
- 3. Return of Zero**
- 4. None of the given

1. In_____each band is reserved for a specific station, and it belongs to the station all the time.

- 1. FDMA**
- 2. CDMA
- 3. TDMA
- 4. CSMA

2. In TCP/IP protocol suite, the process of adding header at each layer on sending side is known as_____.

- 1. Encapsulation
- 2. De-encapsulation
- 3. Packetizing**
- 4. Framing

3. In Block coding we divide our message into blocks of k bits called as_____.

- 1. Code words
- 2. Data words**

3. Passwords

4. Cross words

4. In circuit switched networks we have low efficiency but minimal_____.

1. Delay

2. Speed

3. Throughput

4. Errors

5. Error correction is more _____ than the error detection.

1. Easy

2. Useless

3. Informal

4. Difficult

6. AM signal requires _____ the BW of original signal

1. Twice

2. Half

3. Same

4. Three Times

7. Asynchronous transmission is_____.

1. Slow

2. Fast

3. Costly

4. None effective

8. The Internet is_____.

1. software for sending e-mail around the world
2. a government-owned agency that links computers
3. a global network of computers networks
4. a specialised form of local area network

9. Time division multiplexing is used in_____systems.

1. Analog
2. Hybrid
3. Digital
4. Automated

10. Radio wave transmission utilizes_____different types of propagation.

1. Two
2. Three
3. Four
4. Five

11. Checksum is an error-detection technique that can be applied to a message of_____length.

1. Exactly 4
2. Exactly 8

3. Exactly 32
4. Any
12. At the Application layer, object/information is in the form of _____.
1. Message
 2. Packet
 3. Segment
 4. Frame
13. POST stand for _____?
1. Plain Old Telephone Set
 2. Plain Old Time Sharing
 3. Plain Old Telephone System
 4. Plain Old Telephone Service
14. Which of the following statement is correct?
1. Digital signals are less prone to Noise
 2. Digital signals are highly prone to Noise
 3. Analog Signals are less prone to Noise
 4. Digital signals are not affected by Noise
15. In selective-reject ARQ, only the specific damaged or lost frame is _____.
1. Retransmitted
 2. Forwarded

3. Selected
4. Rejected
16. In ASK correct formula for calculating the bandwidth is as _____.
1. $B=(1+d)S$
 2. $B=(1*d)S$
 3. $B=(d-1)S$
 4. $B=(d-5)S$
17. Encoding digital data into digital signals is called _____.
1. Analog-to-Digital Conversion
 2. Digital-to-Digital Conversion
 3. Digital-to-Analog Conversion
 4. None of the given
18. If the Hamming distance between sent and received code word is _____, then it shows that received data is corrupted.
1. $\neq 0$
 2. $= 0$
 3. $\neq 1$
 4. $= 1$
19. Quadrature Amplitude Modulation (QAM) is the combination of _____ and _____.
1. FSK, PSK

2. PSK, FSK
 3. ASK, PSK
 4. ASK, FSK
20. **PSK is susceptible to the noise degradation.**
1. True
 2. False
21. **___works with topologies in which one device is designated as a primary station and the other devices are secondary stations.**
1. Reservation
 2. **Polling**
 3. Token
 4. None of the given
22. **No matter whether the link is dedicated or broadcast, data link control (DLC) layer provides services between_____.**
1. Virtual LANs
 2. **Two adjacent nodes**
 3. Ethernet links
 4. Source and destination PCs
23. **CDMA stands for_____.**
1. Carrier Division Multiple Access
 2. Carrier Data Multiple Access

3. Code Data Multiple Access

4. Code Division Multiple Access

24. If users must take turns using the link, then it is called Spatial Sharing.

1. true

2. false

25. GO-BACK-N protocol is one of the _____ layer protocols, which is used to control the flow of data during the transmission.

1. Physical

2. Network

3. Presentation

4. Data link

26. In _____ a dedicated communication path is established between two stations through the nodes of the network

1. Circuit switching

2. Packet switching

27. Signals travel through fiber optic cable are in the form of _____.

1. Light

2. Bits

3. Electromagnetic

4. Bytes

28. _____ contains a repeater.

1. Active hub

2. Passive hub

29. In frequency shift keying, _____ remain(s) constant.

1. Frequency

2. Amplitude

3. Both Amplitude and Phase

4. Both Phase and Frequency

30. _____ is/are un-reliable protocol(s).

1. IP and TCP

2. UDP

3. IP and UDP

4. TCP

31. _____ is not a function of Data Link Control.

1. Framing

2. Modulation

3. Error Control

4. Flow Control

32. Each computer on the Internet has a unique numeric address called a(n) _____.

1. domain address

2. protocol address

3. IP address

4. Web address

33. dB is negative if a signal is _____.

1. Attenuated

2. Distorted

3. Amplified

4. Noisy

34. We can have _____ different groups with a 4 bit block.

1. 4

2. 8

3. 16

4. 32

35. Line Configuration refers to the way two or more devices attach to a _____.

1. Path

2. Circuit

3. Link

4. Router

36. Error detection and correction are the services provided by _____ layer.

1. Data link layer

2. Application layer
3. Physical layer
4. Session layer

37. Which of the following sublayer resolves the contention for the shared media.

1. MAC
2. LLC
3. Physical
4. None of the given.

38. There are _____ basic categories of multiplexing.

1. 2
2. 3
3. 7
4. 5

39. Low pass channel has _____ bandwidth between two stations.

1. Dedicated
2. Shared
3. Multiplexed
4. Infinite

40. BSC supports _____ transmission using _____ flow control.

1. Full-duplex, Stop and wait ARQ
2. Half-duplex, Selective reject ARQ
3. Full-duplex, Selective reject ARQ
4. Half-duplex, Stop and wait ARQ

41. Quantization is a method of assigning _____ values in a specific range to sampled instances.

1. Negative
2. Positive
3. Binary
4. Integral

42. _____ encoding is almost obsolete today

1. Bipolar
2. Unipolar
3. Polar
4. None of the given

43. YMODEM has _____ Byte of data unit.

1. 1024
2. 256
3. 128
4. 512

44. Like Data link layer, _____ is also responsible for Flow control.

1. Transport Layer

2. Session Layer

45. In CRC there is no error if the remainder at the receiver is _____.

1. Equal to the remainder at the sender

2. Zero

3. Nonzero

4. The quotient at the sender

46. _____ is normally used where speed is priority in data transfer.

1. Serial data transmission

2. Hybrid data transmission

3. Parallel data transmission

4. Both parallel and serial data transmission

47. In Block coding scheme, number of code words is always _____ data words.

1. Less than

2. Equal to

3. Greater than

4. Same as

48. A portion of the path that carries TX between a given pair of devices is known as _____.

1. Direct Point

2. Bridge
3. Channel
4. Access Point

49. What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 200, 400, 600 and 800?

1. 100
2. 500
3. 300
4. 600

50. _____ is the process of converting digital data to digital signals.

1. CSMA/CD
2. CSMA/CA
3. Line coding
4. Segmentation

51. _____ transmission technology (based on Ethernet) provides a data rate of 1 billion bits per second.

1. Fast Ethernet
2. Standard Ethernet
3. Gigabit Ethernet
4. 10BASE-T Ethernet

52. Congestion Control is a feature of layer(s).

1. Data Link
2. Network
3. Transport
4. **All mentioned**

53. In Pure ALOHA, the vulnerable time is _____ the frame transmission time.

1. Same as
2. **Two times**
3. Three times
4. Four times

54. What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 300, 600, 900 and 1200?

1. 100
2. **900**
3. 300
4. 600

55. _____ signal can take infinite levels of intensity over time.

1. Digital
2. Discrete
3. **Analog**
4. Logical

56. Mr. Asif while sitting in Lahore is talking with his friend in Dubai through Skype is an example of _____.

1. Local Area Network
2. **Metropolitan Area Network**
3. Wide Area Network
4. Home Based Network

57. Which layer is concerned with the syntax and semantics of info exchange between two systems?

1. Physical
2. Application
3. Session
4. **Presentation**

58. Time domain shows changes in _____.

1. time w.r.t signal amplitude
2. **signal amplitude w.r.t time**
3. time
4. All of the given

59. In ASK, Noise usually affects the _____.

1. **Amplitude**
2. Phase
3. Sample

4. Frequency
60. In analog transmission, the base signal is called the_____.
1. Carrier Signal
 2. Analog signal
 3. Digital signal
 4. Modulated signal
61. Which of the following is most affected by noise?
1. PSK
 2. ASK
 3. FSK
 4. QAM
62. In Synchronous TDM, Time slots are not same in number as input devices.
1. True
 2. False
63. The flag in PPP is a byte that needs to be_____whenever it appears in the data section of the frame.
1. Duplicated
 2. Blocked
 3. Cleaned
 4. Escaped

64. Guard bands are used in _____ multiplexing technique, to avoid overlapping of frequency bands assigned to each user.

1. PDM
2. CSMA
3. TDM
4. FDM

65. DLC in Data Link Layer stands for _____.

1. Divide Line Communication
2. Data Line Code
3. Data Link Control
4. Demand Link Coordination

66. In Pulse Code Modulation (PCM), the sampling is dependent on _____.

1. Time
2. Amplitude
3. Frequency
4. Signal Rate

67. Digital signals are referred to be _____.

1. Continuous
2. Discrete
3. All of the given

4. None of the given

68. Two or more computers connected so that they can communicate with each other and share information is called a _____.

1. satellite
2. protocol
3. broadcast
- 4. network**

69. Which error detection method uses one's complement arithmetic.

1. Cyclic Redundancy Check (CRC)
2. Longitudinal Red Check (LRC)
- 3. Checksum**
4. Simple Parity Check

70. WDM stands for _____.

- 1. Wave Division Multiplexing**
2. Wrong Division Multiplexing
3. Well Division Multiplexing
4. West Division Multiplexing

71. Analog refers to something that is continuous in _____.

- 1. Time**
2. Space
3. Frequency

4. None of the given

72. What is the frequency of a sine wave if it completes one cycle in 5 seconds?

1. 0.5

2. 0.6

3. 0.3

4. 0.2

73. In an analog hierarchy to carry voice channels, a group can carry _____ voice channels.

1. 60

2. 12

3. 20

4. 10

74. STP is more expensive than UTP and is less susceptible to noise.

1. True

2. False

75. Switch is a network device which operates on the _____ layer of the TCP/IP protocol suite.

1. Application

2. Transport

3. Presentation

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4. Data link

76. ___ are used to exchange session management and control information between connected devices.

- 1. I-frames**
- 2. U-frames**
- 3. S-frames**
- 4. N-frames**

77. The message 1110 sent by a source is received by a destination as 1011.

This is _____ type of error.

- 1. Burst**
- 2. Single-Bit**
- 3. Uni-Bit**
- 4. Hamming**

Question No: 1

Ethernet Address is of----- bytes

- 4
- **6**
- 8
- 12

Question No: 2

In Pure ALOHA, the vulnerable time is -----the frame transmission time.

- Same as
- **Two times**
- Three times
- Four times

Question No: 3

_____ was used as a medium in 10 Base-T

- Thick coaxial Cable
- Think coaxial Cable
- **Twisted Pair Cable**
- Two Mode For Cable

Question No: 4

When data are transmitted from device A to device B, the header from A's layer 4 is read by B's

_____ layer.

- **physical**
- transport
- application
- none of the given

Question No: 5

In _____ encoding, we use three levels: positive, zero, and negative.

- polar
- nonpolar
- mmultilevel
- **bipolar**

Question No: 6

A periodic signal completes one cycle in 0.001 s. What is the frequency?

- 1 Hz
- **100 Hz**
- 1 KHz
- 1 MHz

Question No: 7

There are _____ basic modes for the propagation of light in fiber-optic cables.

- 3
- 4
- 5
- 2

Question No: 8

A parabolic dish antenna is a(n) _____ antenna.

- Omnidirectional
- Bidirectional
- Horn
- Unidirectional

Question No: 9

Zero crossing bandwidth is also called as equivalent noise bandwidth.

- True
- False

Question No: 10

During the transmission, transmitting device produces a high frequency signal which acts as a basis for the transmission. This base signal is called__.

- Analog signal
 - Digital signal
 - Modulated signal
 - Carrier signal
- Not Confirm

Question No 11:

There are _____ types of serial transmission:

- 1
- 2
- 3
- 4

Question No 12:

Synchronous transmission have_____.

- a start bit
- a stop bit
- gaps between bytes
- none of the given

Question No 13

Using _____ encoding scheme in digital transmission, we represent 0 by zero voltage level and represent 1 by any positive voltage level.

- Polar
- AMI
- Bipolar
- Unipolar

Question No 14

_____ is the formula for finding number of bits sent per level.

- Log L
- Log2
- Log2*2
- Log2 L

Question No 15

Which of the following is an example of ITU-T modem standards:?

- T-series
- X-series
- N-series
- V-series

Question No 16

The maximum data rate in the uploading direction is still _____.

- 26.6 Kbps
- 33.6 Kbps
- 36.6 Kbps
- 46.6 Kbps

Question No 16

In _____ each station is forced to send a frame only at beginning of time slot

- Pure ALOHA
- Slotted ALOHA
- Fragmented ALOHA
- None of the given

Question 17

In _____ each station sends a frame whenever it has a frame to send

- Pure ALOHA
- Slotted ALOHA
- Fragmented ALOHA
- None of the given

Question No 18

The data rate of a T-1 line is _____.

- 1.544 Gbps
- 2.544 Mbps
- 1.544 Kbps
- 1.544 Mbps

Question 19

Circuit switched networks have _____ efficiency but _____ delay compared to the packet switched networks.

- high, minimal

- high, high
- low, minimal

Question No 20

Category 5 UTP cable is used for data transmission of upto_.

- 100 Mbps
- 200 Mbps
- 250 Mbps
- 400 Mbps

Question No 21

All of popular Fiber optic connectors are ____shaped.

- Conical
- Barrel
- Circular
- Rectangular

Question No 22

SONET stands for _____

- synchronous optical network
- synchronous operational network
- stream optical network
- shell operational network

Question No 23

Fast Ethernet has a data rate of _____Mbps.

- 10
- 100
- 1000
- 10,000

Question No 24

The VLF and LF bands use _____propagation for communications.

- Ground
- Sky
- Line of sight
- Space

Question No 25

In Bit-oriented framing, _____bit pattern is used as a flag to define beginning and end of a frame.

- $(11111111)_2$
- $(10000001)_2$
- Single
- $(01111110)_2$

Question No 26

Traditionally, _____protocols have been defined for the data-link layer to deal with flow

and error control.

- Two
- Three

- Five
- Four

Question No 27

Multiplexing is the set of techniques that allows simultaneous TX of multiple signals across _____ data link

- Single
- Multi
- Single and Multi
- none of the given

Question No 28

Need for Addressing makes Asynchronous TDM inefficient for bit or byte.

- Interleaving
- Addition
- Substraction
- None of the given

Question No 29

_____ takes data from one high speed line and breaks it into portions.

- ▶ Multiplexing
- ▶ Inverse multiplexing
- ▶ Inverse subtraction
- ▶ Inverse addition

Question No 30

For Carrier Sense Multiple Access/Collision Detection (CSMA/CD) we need a restriction on the

- Collision Size
- Signal Size
- Frame Size
- Station Size

Question No 31:

In Y-MODEM Multiple files can be sent simultaneously

- ▶ True
- ▶ False

Question No 32

At the data-link layer, _____ protocol is designed to handle both flow and error control,

but communication is one frame at a time.

- Simple
- Go-Back-N
- Selective-Repeat
- Stop-and-Wait

Question No 33

The PPP protocol uses _____, which is responsible for establishing, maintaining, configuring, and terminating links.

- Internet Protocol Control Protocol (IPCP)
- Challenge Handshake Authentication Protocol (CHAP)
- Network Control Protocols (NCPs)
- **Link Control Protocol (LCP)**

Question No 34

Aloha, CSMA/CD and CSMA/CA are part of _____.

- Controlled Access protocol
- Channelization protocol
- Mutual access protocol
- Random Access protocol**

Question No 35

Which of the following is an example of ITU-T modem standards:

- **T-series**
- X-series
- N-series
- V-series

Question No 36

The maximum data rate in the uploading direction is still _____.

- 26.6 Kbps
- **33.6 Kbps**
- 36.6 Kbps
- 46.6 Kbps

Question No 37

Category 5 UTP cable is used for data transmission of upto_.

- **100 Mbps**
- 200 Mbps
- 250 Mbps
- 400 Mbps

Question No 38

ALOHA the earliest random access method was developed in early_____.

1950s
1960s

1980s
1970s

Question No 39

In block coding, we divide our message into blocks, each of _____bits; and add _____redundant

bits to each block to make the codewords of

_____bit

length.m, r, c

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n, k, r
n, r, k
k, r, n

Question No 40

All of popular Fiber optic connectors are ____ shaped.

- Conical
- Barrel
- Circular
- Rectangular

Question No 41

MAC stand for _____

- Media Access Controller
- Main Access Control
- Mandatory Access Control
- Media Access Control

Question No 42

Which one of the following is a Random-Access protocol

- CDMA/CA
- ALOHA
- CDMA/CD
- TDMA

Question No 43

The VLF and LF bands use _____ propagation for communications.

- Ground
- Sky
- Line of sight
- Space

Question No 44

Multiplexing is the set of techniques that allows simultaneous TX of multiple signals across

_____ data link

- Single
- Multi
- Single and Multi
- none of the given

Question No 45

Need for Addressing makes Asynchronous TDM inefficient for bit or byte.

- Interleaving
- Addition
- Subtraction
- None of the given

Question No 46

CDMA stands for _____

- Carrier Data Multiple Access
- Code Division Multiple Access
- Code Data Multiple Access
- Carrier Division Multiple Access

Question No 47

In Ethernet, the source address field in the MAC frame is the _____ address.

- Sender's logical
- Previous station's physical
- Next destination's physical
- Sender's service port

Question No 48

_____ takes data from one high speed line and breaks it into portions.

- Multiplexing
- Inverse multiplexing
- Inverse subtraction
- Inverse addition

Question No 49

A traditional telephone line has a bandwidth of _____.

- ▶ 2000Hz
- ▶ 4000 Hz
- ▶ 2000 MHz
- ▶ 4000 MHz

Question No 50

To detect five errors, the Hamming distance between each pair of codewords should be at least ____.

- 4
- 5
- 11
- 6

Question No 51

For FEC using XOR operation, we divide a packet into N chunks, and need to send _____ chunks to the receiver side.

- N
- 2N
- $(N+1)/2$
- N+1

Question No 52

If the ASCII character H is sent and the character I is received, what type of error is this?

- Single-bit
- Multiple-bit
- Burst
- Recoverable

Question No 53

At the CRC generator, _____ added to the data unit after the division process

- 0s are
- 1s are
- The polynomial
- The CRC remainder is

Question No 54

Error control in the data link layer is based on _____

- automatic repeat request
- automatic repeat acknowledgment
- automatic send acknowledgment

Question No 55

In a technique called, compounding High-and-Low Resolution

► **CRC** Not

Packets, the low-resolution section in the first packet is__.

- Parity code
- Empty

Question No 56

In line discipline the initiator first transmits a frame called an _____

- Enquiry
- Acknowledgment
- **NAK**
- Reques

Question No 57

Primary device uses _____ to receive transmission from the secondary devices.

► **ACK** Not

- ENQ
- POLL

Question N0 58

Token Bus has no commercial application in data communications

- **True**
- False

Question No 59

HDLC is an acronym for_____.

- High-duplex line communication
- **High-level data link control**
- Half-duplex digital link combination
- Host double-level circuit

Question No 59

The BNC-T connector is a T-shaped device with__ports

- Three
- **Two**
- Four
- Five

Question No 60

Each station in the Token Ring regenerates the frame.

- **True**
- False

Question No 61

Repeater works on _____ layer.

- Data Link
- Physical
- **Network**
- Application

Question No 62

Trunks are transmission media such as _____ that handle the telephone to the nearest end office.

- Satellite links
- **Twisted-pair & Fiber-optic**
- Twisted-pair
- Fiber-optic

Question No 63

At the data-link layer, _____ protocol is designed to handle both flow and error control, but communication is one frame at a time.

- Simple
- Go-Back-N
- Selective-Repeat
- **Stop-and-Wait**

Question No 64

The PPP protocol uses _____, which is responsible for establishing, maintaining, configuring, and terminating links.

- Internet Protocol Control Protocol (IPCP)
- Challenge Handshake Authentication Protocol (CHAP)
- Network Control Protocols (NCPs)
- **Link Control Protocol (LCP)**

Question No 65

The original Ethernet technology with the data rate of 10 Mbps is called?

- **Standard Ethernet**
- Fast Ethernet
- Gigabit Ethernet
- 10 Gigabit Ethernet

Question No 66

Normally the value of Kmax in pure ALOHA is ____.

- 10
- **15**
- 25
- 30

Question No 67

Which one of the following is not a Channelization Protocol?

- **CSMA**
- CDMA

➤ FDMA

➤ TDMA

Question No 68

In CSMA/CA _____ is the amount of time divided into slots.

➤ Contention Window

➤ Interframe Window

➤ Fragmented Window

➤ Collided Window

Question No 69

Aloha, CSMA/CD and CSMA/CA are part of _____.

Controlled Access protocol

Channelization protocol

Mutual access protocol

Random Access protocol

Question No 70

ALOHA the earliest random access method was developed in early ____.

• 1950s

• 1960s

• 1980s

• 1970s

Question No 71

For FEC using XOR operation, we divide a packet into N chunks, and need to send _____ chunks to the receiver side.

N

2N

$(N+1)/2$

2N+1

Question No 71

The _____ layer is responsible for the creation and delivery of a frame to another node, along the link.

➤ Application

➤ Network

➤ Physical

➤ Data-link

Question No 72

IP addresses are the identifiers at the _____ layer.

Physical

Data-link

Transport

➤ Network

Question No 73

Each host or each interface of a router is assigned _____ address.

➤ A multicast

➤ A broadcast

➤ Both multicast or broadcast

- A unicast

Question No 74

The ARP protocol is one of the auxiliary protocols defined in the _layer, which maps an IP address to a logical-link address.

- Physical
- Data-link
- Transport
- Network

Question No 75

Following are the examples of connecting devices EXCEPT ____.

- Link Layer Switches
- Hub
- Routers
- Wire

Question No 76

_____ eliminates collision and connect heterogeneous devices.

- Hub
- Repeater
- Bridge
- Switch

Question No 77

_____ creates looping problem in learning switch.

- Unicast
- Multicast
- None of the Given
- Broadcast

Question No 78

The term FTTC stands for ____.

Front Technology Terminal

Connection Free Technology

Transmission Connection

Fiber Terminal Transmission

Commission Fiber to the Curb

Question No 79

Multiple-access Protocol is divided into ____.

- Two categories
- Four categories
- Five categories

Three
categories

Azi.kkhan

Question No 80

The original Ethernet technology with the data rate of 10 Mbps is called ___ Ethernet.

- Fas
- t
- Gigab
- it
- 10
- Gigabit
- Standard**

Question No 81

CSMA/CD is not used in _____ Ethernet.

- Standa
- rdFast
- Gigabit
- 10 Gigabit**

Question No 82

The amplitude of a digital signal depends upon the ___ to represent a bit.

- Phase
- Waveleng
- th
- Bandwidt
- h Voltage**

Question No 83

In IEEE 802.11, a station with ___ mobility can move from one BSS to another, but the movement is confined inside one ESS.

- None of the choices are
- correctNo-transition
- ESS-transition**
- BSS-transition

Question No 84

In IEEE 802.11, communication between two stations in two different BSSs usually occurs via two ____.

- BSS
- s
- ESS
- s
- None of the
- given**Apis**

Question No 89

A property of a signal where it is being received by receiver after reflection from different intermediate objects (Building, iron polls, walls) is called ____

- **Attenuation**
- Interference

- Error
- Multipath Propagation

Question No 90

_____medium is used in wireless LANs for transportation of data.

Copper pair cable

Azizkhan

Optical fiber
cable
Coaxial
cable

Air

Question No 91

In context of control signaling network, what does STP stand for ____

- Shielded twisted pair
- Signaling transmission point
- Signal transportation procedure
- **Signal transfer point**

Question No 92

POP stands for _____

- Presentation of points
- Points of points
- Points of packets
- **Points of Presence**

Question No 93

What does LATA stand for _____

- Local Area Transmission Arena
- Local-Access Transit Access
- Local-Areas Transport Areas
- **Local-Access Transport Areas**

Question No 94

Unipolar encoding scheme is not supported over the medium having _____.

- The capability of handling DC component
- Low speed
- High speed
- **No capability of handling DC component**

Question No 95

The amplitude of a digital signal depends upon the _____ to represent a bit.

- ▶ Phase
 - ▶ Wavelength
 - ▶ Bandwidth
 - ▶ **Voltage**

Question No 96

Transmission Control Protocol is a(n) _____ layer protocol

- Application
- Network
- Session
- **Transport**

Question No 97

In _____ each station is allotted a time slot during which it can send data

- NDMA
- CDMA
- TDMA
- FDMA

Question No 98

In _____ collisions are avoided by deferring transmission even if the channel is idle

- The Slotted Frame
- The Deployment Frame
- The Contention Window
- The Interframe Space

Question No 99

In the context of Ethernet Implementations, maximum medium length of 10 Base-F was _____ meters.

- 500
- 1000
- 2000
- 150

Question No 100

Collisions in CSMA/CA are avoided through _____

- The Interframe Space
- The Contention Window
- Acknowledgments
- All of the given

1. The extra bits added with the original data for error detection/correction are called

- ▶ Process IDs
- ▶ Redundant bits
- ▶ Fault bits
- ▶ Port Number

2. _____ are used to exchange session management and control between connected devices.

- ▶ U-frames
- ▶ N-frames
- ▶ S-frames
- ▶ I-frames

3. The message 1110 sent by source is received by a destination as 1011. This is _____ types of err.

- ▶ Hamming
- ▶ Single-Bit
- ▶ Uni-Bit
- ▶ Brust

4. In Block code we divide our message into blocks of K bits called as _____

- ▶ Password
- ▶ Code word

- ▶ Data words
 - ▶ Cross words
5. DLC in Data Link Layer stand for_____
- ▶ Demand Link coordination
 - ▶ Data Line Code
 - ▶ Data Link Control
 - ▶ Divide Line Communication
6. _____ is responsible for governing node to node Communication.
- ▶ Data Link Layer
 - ▶ Session Layer
 - ▶ Presentation Layer
 - ▶ Application Layer
7. Go-Back-N protocol is one of the _____ Layer protocol, which is used to control the flow of data during the transmission
- ▶ Presentation
 - ▶ Network
 - ▶ Data Link
 - ▶ Physical
8. Error correction is more _____ than the error detection.
- ▶ Useless
 - ▶ Informal
 - ▶ Difficult
 - ▶ Easy
9. Stop-and-Wait protocol is used for _____
- ▶ Flow control Only
 - ▶ Neither flow nor error control
 - ▶ Both flow and error control
10. _____ is not a function of Data Link control
- ▶ Framing
 - ▶ Error Control
 - ▶ Flow Control
 - ▶ Modulation
11. Payload field of the PPP frame is of variable size and it can carry maximum Data up to _____
- ▶ 10 Bytes
 - ▶ 500 Bytes
 - ▶ 100 Bytes
 - ▶ 1500 Bytes
12. In Block Coding Scheme, number of Code word is always _____ data words.
- ▶ Equal to
 - ▶ same as
 - ▶ Greater Than
 - ▶ Less than

13. High-Level Data Link Control(HDLC) defines __types of frames to deal with the transfer modes.

- ▶ Three
- ▶ Two
- ▶ Four
- ▶ Six

14. Congestion control is a feature of _____layer(S)

- ▶ Transport
- ▶ All Mentioned
- ▶ Network
- ▶ Data Link

15. Error detection and correction are the services provided by _____ Layer.

- ▶ Physical Layer
- ▶ Application Layer
- ▶ Data Link Layer
- ▶ Session Layer

16. IN Linear Block codes, __of two valid code words creates another valid code word.

- ▶ AND
- ▶ OR
- ▶ XOR
- ▶ NOR

17. Stop-and-Wait protocol is used for _____

- ▶ Bothe flow and error control
- ▶ Error control Only
- ▶ Flow control Only
- ▶ Nether flow nor error control

18. CHAP is an Authentication protocol, which uses _____process to authenticate user information.

- ▶ Four way
- ▶ Three Way
- ▶ Two Way
- ▶ Five way

19. Data link layer is divided into _____sub Layer

- ▶ Seven
- ▶ Two
- ▶ Five
- ▶ Three

20. If we need to correct a single error in an 8-bit data unit, we need to consider _____possible error location.

- ▶ 16
- ▶ 2
- ▶ 8
- ▶ 4

21. _____ is not a function of Data Link Control.

- ▶ Framing
- ▶ **Modulation**
- ▶ Flow control
- ▶ Error control

22. The _____ is the number of single elements sent per unit time.

- ▶ **Single rate**
- ▶ Period
- ▶ Bit rate
- ▶ Data rate

23. In ASK, both _____ and _____ remain constant.

- ▶ Amplitude, Frequency
- ▶ Amplitude, phase
- ▶ **Frequency, Phase**
- ▶ Amplitude, Samples

24. In an analog hierarchy to carry voice channels, a group can carry _____ **voice channels**.

60

12

20

10

25. Which of the following sublayer resolves the contention for the shared media.

- ▶ **MAC**
- ▶ LLC
- ▶ Physical
- ▶ None of the given.

26. What is the period of a Sine wave having frequency of 5 Hz?

0.2

0.1

0.5

0.3

27. In CRC there is no error if the remainder at the receiver is _____.

- ▶ Equal to the remainder at the sender
- ▶ **Zero**
- ▶ Nonzero
- ▶ The quotient at the sender

28. Analog To Analog Conversion Methods include _____.

- ▶ **AM, PM and FM**
- ▶ AM, FSK and PM
- ▶ AM, PM and QAM
- ▶ None of the given

29. Signals travel through fiber optic cable are in the form of _____.

- ▶ **Light**
- ▶ Bits
- ▶ Electromagnetic
- ▶ Bytes

30.

If the Hamming distance between sent and received code word is _____, **then it shows that received data is corrupted.**

- ▶ **≠ 0**
- ▶ = 0
- ▶ ≠ 1
- ▶ = 1

31. The flag in PPP is a byte that needs to be _____ whenever it appears in the data section of the frame.

- ▶ Duplicated
- ▶ Blocked
- ▶ Cleaned
- ▶ **Escaped**

32. _____ is normally used where speed is priority in data transfer.

- ▶ **Serial data transmission**
- ▶ Hybrid data transmission
- ▶ Parallel data transmission
- ▶ Both parallel and serial data transmission

33. Baud rate is greater than or equal to the bit rate.

- ▶ True
- ▶ **False**

34. _____ is an Authentication Protocol, which is used by the PPP to authenticate passwords.

- ▶ CHAP
- ▶ **PAP**
- ▶ LCP
- ▶ ICMP

35. In Pure ALOHA, the vulnerable time is _____ the frame transmission time.

- ▶ Same as
- ▶ **Two times**
- ▶ Three times
- ▶ Four times

36. A broadcast signal is received by the orbiting satellites which rebroadcasts _.

- ▶ to the resender
- ▶ to the sender
- ▶ to the atmosphere
- ▶ **none of the given**

37. In TCP/IP protocol suite, the process of adding header at each layer on sending side is known as _____.

- ▶ Encapsulation
 - ▶ De-encapsulation
 - ▶ **Packetizing**
 - ▶ Framing
38. By using _____ system, change in the file contents during file transmission can be avoided.
- ▶ Routing
 - ▶ Flow Control
 - ▶ **Error Detection and Correction**
 - ▶ Congestion Control
39. Which of the following consist of just one redundant bit per data unit?
- ▶ Two dimensional Parity check.
 - ▶ CRC
 - ▶ **Simple Parity Check**
 - ▶ Checksum
40. If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause _____ **number of bits being impacted by the noise.**
- ▶ Constant
 - ▶ **More**
 - ▶ Less
 - ▶ Zero
41. The extra bits added with the original data for error detection/correction are called _____.
- ▶ Fault bits
 - ▶ Port numbers
 - ▶ **Redundant bits**
 - ▶ Process IDs
42. Flow control is needed to prevent _____.
- ▶ Bit errors
 - ▶ Overflow of the sender buffer
 - ▶ **Overflow of the receiver buffer**
 - ▶ Collision between sender and receiver
43. The inversion of the level represents a 1 bit in _____ encoding
- ▶ **NRZ-I**
 - ▶ NRZ-L
 - ▶ RZ
 - ▶ Manchester
44. The term _____ means that only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.
- ▶ Packet Error
 - ▶ Burst Error
 - ▶ **Single Bit Error**
 - ▶ Character Error

45. A sine wave is defined by _____ characteristics.

- ▶ 2
- ▶ **3**
- ▶ 4
- ▶ 5

46. Quadrature Amplitude Modulation (QAM) is the combination of _____ and _____.

- ▶ **FSK, PSK**
- ▶ PSK, FSK
- ▶ ASK, PSK
- ▶ ASK, FSK

47. If we need to correct a single error in an 8-bit data unit, we need to consider _____ possible error locations.

• **16**

- 2
- 4
- 8

48. To calculate the data rate for noisy channel _____ formula is used.

- ▶ **Shannon**
- ▶ Nyquist
- ▶ Propagation
- ▶ Greedy

49. GO-BACK-N protocol is one of the _____ layer protocols, which is used to control the flow of data during the transmission.

- ▶ Physical
- ▶ **Network**
- ▶ Presentation
- ▶ Data link

50. A _____ modem is used to connect two DTEs directly.

- ▶ **NULL**
- ▶ Cable
- ▶ DSL
- ▶ None of the given

51. Time division multiplexing is used in _____ systems.

- ▶ Analog
- ▶ Hybrid
- ▶ **Digital**
- ▶ Automated

52. In OSI model headers are added at layer _____.

- ▶ 1,2,3 only
- ▶ 6,5,7 only
- ▶ 5,4,3 only

▶ **6,5,4,3,2 only**

53. Line Configuration refers to the way two or more devices attach to a_____.

- ▶ Path
- ▶ Circuit
- ▶ **Link**
- ▶ Router

54. A sine wave is_____.

- ▶ periodic and discrete
- ▶ aperiodic and discrete
- ▶ **periodic and continuous**
- ▶ aperiodic and continuous

55. The logical connection between the peer layers is_____connection.

- ▶ Physical
- ▶ Direct
- ▶ Indirect
- ▶ **Tangible**

56. What is the frequency of a sine wave if it completes one cycle in 2 seconds?

0.5

0.6

0.3

0.2

57. In Synchronous TDM, Time slots are not same in number as input devices.

- ▶ True
- ▶ **False**

58. _____category of coaxial cable is used for thin Ethernet.

- ▶ **RG-58**
- ▶ RG-59
- ▶ RG-1
- ▶ RG-47

59. Radio wave transmission utilizes_____different types of propagation.

- ▶ Two
- ▶ Three
- ▶ Four
- ▶ **Five**

60. BSC supports_____transmission using _____
flow control.

- ▶ Full-duplex, Stop and wait ARQ
- ▶ Half-duplex, Selective reject ARQ
- ▶ Full-duplex, Selective reject ARQ
- ▶ **Half-duplex, Stop and wait ARQ**

61. In selective-reject ARQ, only the specific damaged or lost frame is_.

- ▶ **Retransmitted**
- ▶ Forwarded
- ▶ Selected

- ▶ Rejected
62. Error detection and correction are the services provided by _____ layer.
- ▶ **Data link layer**
 - ▶ Application layer
 - ▶ Physical layer
 - ▶ Session layer
63. The _____ generates the data and passes it along with any control information to a _____.
- ▶ **DTE, DCE**
 - ▶ DCE, DTE
 - ▶ DTE, DTE
 - ▶ DCE, DCE
64. Analog refers to something that is continuous in _____.
- ▶ **Time**
 - ▶ Space
 - ▶ Frequency
 - ▶ None of the given
65. Time domain shows changes in _____.
- ▶ time w.r.t signal amplitude
 - ▶ **signal amplitude w.r.t time**
 - ▶ time
 - ▶ All of the given
66. Mr. Asif while sitting in Lahore is talking with his friend in Dubai through Skype is an example of _____.
- ▶ Local Area Network
 - ▶ **Metropolitan Area Network**
 - ▶ Wide Area Network
 - ▶ Home Based Network
67. Which level of the TCP/IP reference model routes data/information across a network channel?
- ▶ Application Layer
 - ▶ Data Link Layer
 - ▶ Transport Layer
 - ▶ **Network Layer**
68. There are _____ types of serial transmission.
- 3
 - 4
 - **2**
 - None of the given
69. Latency is made of _____ components.
- ▶ **Two**
 - ▶ Three

- ▶ Four
 - ▶ Five
70. Like 10 Base 5, 10 Base 2 is a _____ topology LAN.
- ▶ Ring
 - ▶ Mesh
 - ▶ Star
 - ▶ **Bus**
71. _____ is the rate of change with respect to time.
- a. Voltage
 - b. Time
 - c. Frequency
 - d. **Amplitude**
72. Two or more computers connected so that they can communicate with each other and share information is called a _____.
- ▶ satellite
 - ▶ protocol
 - ▶ broadcast
 - ▶ **network**
73. Which error detection method consists of a parity bit for each unit as well as an entire data unit of parity bits?
- ▶ Checksum
 - ▶ Cyclic Redundancy Check (CRC)
 - ▶ **Longitudinal Red Check (LRC)**
 - ▶ Vertical Redundancy Check (VRC)
74. Central Hub in a Tree is an _____.
- ▶ **Active hub**
 - ▶ passive hub
75. Time domain plot shows changes in signal phase with respect to time.
- ▶ true
 - ▶ **false**
76. _____ is an Authentication Protocol, which uses two-step process to authenticate user information.
- ▶ CHAP
 - ▶ LCP
 - ▶ **PAP**
 - ▶ ICMP
77. CHAP is an Authentication Protocol, which uses _____ process to authenticate user information.
- ▶ Two-way
 - ▶ **Three-way**
 - ▶ Four-way
 - ▶ Five-way

78. No matter whether the link is dedicated or broadcast, data link control (DLC) layer provides services between _____.
▶ Virtual LANs
▶ **Two adjacent nodes**
▶ Ethernet links
▶ Source and destination PCs
79. Encoding digital data into digital signals is called _____.
▶ Analog-to-Digital Conversion
▶ **Digital-to-Digital Conversion**
▶ Digital-to-Analog Conversion
▶ None of the given
80. _____ encoding is almost obsolete today
▶ Bipolar
▶ **Unipolar**
▶ Polar
▶ None of the given
81. The Internet is _____.
▶ software for sending e-mail around the world
▶ a government-owned agency that links computers
▶ **a global network of computers networks**
▶ a specialised form of local area network
82. PSK is susceptible to the noise degradation.
▶ True
▶ **False**
83. Low pass channel has _____ bandwidth between two stations.
▶ **Dedicated**
▶ Shared
▶ Multiplexed
▶ Infinite
84. In ASK, Noise usually affects the _____.
▶ **Amplitude**
▶ Phase
▶ Sample
▶ Frequency
85. The flag in PPP is a byte that needs to be _____ whenever it appears in the data section of the frame.
▶ Duplicated
▶ Blocked
▶ Cleaned
▶ **Escaped**
86. Which of the following is not a characteristic of a sine wave?
▶ Amplitude
▶ **Segmentation**
▶ Phase

- ▶ Frequency

87. In statistical time division multiplexing, number of slots in frame are less than

-
- ▶ Output data rate
 - ▶ **Input lines**
 - ▶ Input frequency
 - ▶ Multiplexed T lines

88. If a digital signal has "L" number of levels.

_____ is the number of bits required to represent each level.

- ▶ **Log L**
- ▶ Log Base 2 L
- ▶ L Square
- ▶ $\text{Log}(2*L)$

89. _____ is responsible for governing node to node communication.

- ▶ Application Layer
- ▶ Session Layer
- ▶ **Data Link layer**
- ▶ Presentation Layer

90. The original Ethernet technology with the data rate of 10 Mbps is called _____?

- ▶ **Standard Ethernet**
- ▶ Fast Ethernet
- ▶ Gigabit Ethernet
- ▶ 10 Gigabit Ethernet

91. What is the period of a Sine wave having frequency of 2 Hz?

- 0.2
- 0.1
- 0.5**
- 0.3

92. In line discipline after the data transmission, the sending system finishes with an _____

- ▶ **EOT**
- ▶ EKT
- ▶ ENT
- ▶ ESP

93. _____ are used for Multi cast communication, such as radio and Television.

- ▶ Microwaves
- ▶ **Radio Waves**
- ▶ Light waves
- ▶ Infrared Waves

1. Which of the following sublayer resolves the contention for the shared media?

1. **MAC**
2. LLC
3. Physical
4. None of the given.

2. What is the period of a Sine wave having frequency of 5Hz?

1. **0.2**
2. 0.1 3. 0.5
4. **0.3**

3. In CRC there is no error if the remainder at the receiver is.

1. Equal to the remainder at the sender
2. **Zero**
3. Nonzero
4. The quotient at the sender

4. Analog To Analog Conversion Methods include

-
1. **AM, PM and FM**
 2. AM, FSK and PM
 3. AM, PM and QAM
 4. None of the given

5. Signals travel through fiber optic cable are in the form of

-
1. **Light**
 2. Bits
 3. Electromagnetic
 4. Bytes

6. If the Hamming distance between sent and received code word is , then it shows that received data is corrupted.

1. **$\neq 0$**

2. $= 0$ 3. $\neq 1$

4. $= 1$

7. The flag in PPP is a byte that needs to be _____
whenever it appears in the data section of the frame.

1. Duplicated
2. Blocked
3. Cleaned
4. **Escaped**

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8. _____ is normally used where speed is priority in data transfer.
1. **Serial data transmission**
 2. Hybrid data transmission
 3. Parallel data transmission
 4. Both parallel and serial data transmission
9. Baud rate is greater than or equal to the bit rate.
1. True
 2. **False**
10. _____ is an Authentication Protocol, which is used by the PPP to authenticate passwords.
1. CHAP
 2. **PAP**
 3. LCP
 4. ICMP
11. In Pure ALOHA, the vulnerable time is _____ the frame transmission time.
1. Same as
 2. **Two times**
 3. Three times
 4. Four times
12. A broadcast signal is received by the orbiting satellites which rebroadcasts _____.
1. to the resender
 2. to the sender
 3. to the atmosphere
 4. **none of the given**
13. In TCP/IP protocol suite, the process of adding header at each layer on sending side is known as _____.
1. Encapsulation
 2. De-encapsulation
 3. **Packetizing**
 4. Framing
14. By using _____ system, change in the file contents during file transmission can be avoided.
1. Routing
 2. Flow Control
 3. **Error Detection and Correction**
 4. Congestion Control
15. Which of the following consist of just one redundant bit per data unit?
1. Two-dimensional Parity check.

2. CRC
 3. **Simple Parity Check**
 4. Checksum
16. If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause__number of bits being impacted by the noise.
1. Constant
 2. **More**
 3. Less
 4. Zero
17. The extra bits added with the original data for error detection/correction are called_____.
1. Fault bits
 2. Port numbers
 3. **Redundant bits**
 4. Process IDs
18. Flow control is needed to prevent_____.
1. Bit errors
 2. Overflow of the sender buffer
 3. **Overflow of the receiver buffer**
 4. Collision between sender and receiver
19. The inversion of the level represents a 1 bit in _____ encoding
1. **NRZ-I**
 2. NRZ-L
 3. RZ
 4. Manchester
20. The term_____means that only one bit of a givendata unit is changed from 1 to 0 or from 0 to 1.
1. **Packet Error**
 2. Burst Error
 3. **Single Bit Error**
 4. Character Error
21. A sine wave is defined by_____characteristics.
1. 2
 2. **3**
 3. 4
 4. 5
22. Quadrature Amplitude Modulation (QAM) is the combination of__and_____.
1. **FSK, PSK**

2. PSK, FSK
 3. ASK, PSK
 4. ASK, FSK
23. If we need to correct a single error in an 8-bit data unit, we need to consider _____ possible error locations.
1. **16**
 2. 2 3. 4
- 4. 8**
24. To calculate the data rate for noisy channel _____ formula is used.
1. **Shannon**
 2. Nyquist
 3. Propagation
 4. Greedy
25. GO-BACK-N protocol is one of the _____ layer protocols, which is used to control the flow of data during the transmission.
1. Physical
 2. **Network**
 3. Presentation
 4. Data link
26. A _____ modem is used to connect two DTEs directly.
1. **'NULL'**
 2. Cable
 3. DSL
 4. None of the given
27. Time division multiplexing is used in _____ systems.
1. Analog
 2. Hybrid
 3. **Digital**
 4. Automated
28. In OSI model headers are added at layer _____.
1. 1,2,3 only
 2. **6,5,7 only**
 3. 5,4,3 only
 4. **6,5,4,3,2 only**
29. Line Configuration refers to the way two or more devices attach to a .
1. Path
 2. Circuit

3. Link

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4. Router
30. A sine wave is _____.
1. periodic and discrete
2. aperiodic and discrete
3. **periodic and continuous**
4. aperiodic and continuous
31. The logical connection between the peer layers is _____ **connection**.
1. Physical
2. Direct
3. Indirect
4. **Tangible**
32. What is the frequency of a sine wave if it completes one cycle in 2 seconds? 1. **0.5**
2. 0.6
3. 0.3
4. 0.2
33. In Synchronous TDM, Time slots are not same in number as input devices. 1. True
2. **False**
36. _____ **category of coaxial cable is used for thin Ethernet**.
1. **RG-58**
2. RG-59
3. RG-1
4. RG-47
37. Radio wave transmission utilizes _____ different types of propagation. 1. Two
2. Three
3. Four
4. **Five**
38. BSC supports _____ **transmission using _____ flow control**.
1. full-duplex, Stop and wait ARQ
2. Half-duplex, Selective reject ARQ
3. Full-duplex, Selective reject ARQ
4. **Half-duplex, Stop and wait ARQ**
39. In selective-reject ARQ, only the specific damaged or lost frame is
.
1. **Retransmitted**

2. Forwarded
3. Selected

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4. Rejected
40. Error detection and correction are the services provided by ____ layer.
1. **Data link layer**
 2. Application layer
 3. Physical layer
 4. Session layer
41. The _____ generates the data and passes it along with any control information to a _____.
1. **DTE, DCE**
 2. DCE, DTE
 3. DTE, DTE
 4. DCE, DCE
42. Analog refers to something that is continuous in _____.
1. **Time**
 2. Space
 3. Frequency
 4. None of the given
43. Time domain shows changes in _____.
1. time w.r.t signal amplitude
 2. **signal amplitude w.r.t time**
 3. time
 4. All of the given
44. Mr. Asif while sitting in Lahore is talking with his friend in Dubai through Skype is an example of _____.
1. Local Area Network
 2. **Metropolitan Area Network**
 3. Wide Area Network
 4. Home Based Network
45. Which level of the TCP/IP reference model routes data/information across a network channel?
1. Application Layer
 2. Data Link Layer
 3. Transport Layer
 4. **Network Layer**
46. There are _____ types of serial transmission.
1. 3
 2. 4
 3. **2**
 4. **None of the given**
47. Latency is made of _____ components.

1. **Two**
 2. Three
 3. Four
 4. Five
48. Like 10 Base 5, 10 Base 2 is a _____ topology LAN.
1. Ring
 2. Mesh
 3. Star
 4. **Bus**
49. _____ is the rate of change with respect to time.
1. Voltage
 2. Time
 3. Frequency
 4. **Amplitude**
50. Two or more computers connected so that they can communicate with each other and share information is called a__.
1. satellite
 2. protocol
 3. broadcast
 4. **network**
51. Which error detection method consists of a parity bit for each unit as well as an entire data unit of parity bits?
1. Checksum
 2. Cyclic Redundancy Check (CRC)
 3. **Longitudinal Red Check (LRC)**
 4. Vertical Redundancy Check (VRC)
52. Central Hub in a Tree is an _____.
1. **Active hub**
 2. passive hub
53. Time domain plot shows changes in signal phase with respect to time.
1. true
 2. **false**
54. _____ is an Authentication Protocol, which uses a two-step process to authenticate user information.
1. CHAP
 2. LCP
 3. **PAP**
 4. ICMP
55. CHAP is an Authentication Protocol, which uses _____ process to authenticate user information.

1. Two-way
 2. **Three-way**
 3. Four-way
 4. Five-way
56. No matter whether the link is dedicated or broadcast, datalink control (DLC) layer provides services between _____.
1. Virtual LANs
 2. **Two adjacent nodes**
 3. Ethernet links
 4. Source and destination PCs
57. Encoding digital data into digital signals is called _____.
1. Analog-to-Digital Conversion
 2. **Digital-to-Digital Conversion**
 3. Digital-to-Analog Conversion
 4. None of the given
58. _____ encoding is almost obsolete today
1. Bipolar
 2. **Unipolar**
 3. Polar
 4. None of the given
59. The Internet is _____.
1. software for sending e-mail around the world
 2. a government-owned agency that links computers
 3. **a global network of computers networks**
 4. a specialized form of local area network
60. PSK is susceptible to the noise degradation.
1. True
 2. **False**
61. Low pass channel has _____ bandwidth between two stations.
1. **Dedicated**
 2. Shared
 3. Multiplexed
 4. Infinite
62. In ASK, Noise usually affects the _____.
1. **Amplitude**
 2. Phase
 3. Sample
 4. Frequency

63. The flag in PPP is a byte that needs to be _____ whenever it appears in the data section of the frame.
1. Duplicated
 2. Blocked
 3. Cleaned
 4. **Escaped**
64. Which of the following is not a characteristic of a sinewave? 1. Amplitude
2. **Segmentation**
 3. Phase
 4. Frequency
65. In statistical time division multiplexing, number of slots in frame are less than _____.
1. Output data rate
 2. **Input lines**
 3. Input frequency
 4. Multiplexed T lines
66. If a digital signal has "L" number of levels, _____ is the number of bits required to represent each level.
1. **Log L**
 2. Log Base 2 L
 3. L Square
 4. Log(2*L)
67. _____ is responsible for governing node to node communication.
1. Application Layer
 2. Session Layer
 3. **Data Link layer**
 4. Presentation Layer
68. The original Ethernet technology with the data rate of 10 Mbps is called _____?
1. **Standard Ethernet**
 2. Fast Ethernet
 3. Gigabit Ethernet
 4. 10 Gigabit Ethernet
69. What is the period of a Sine wave having frequency of 2 Hz?
1. 0.2
 2. 0.1
 3. **0.5**
 4. 0.3

70. In line discipline after the data transmission, the sending system finishes with an _____ frame.

1. **EOT**
2. EKT
3. ENT
4. ESP

71. Each computer on the Internet has a unique numeric address called a(n) _____.

1. domain address
2. protocol address
3. **IP address**
4. Web address

72. In even parity check code, the value of syndrome is _____ **if the number of 1s is even.**

1. 1
2. 2
3. **0**
4. 4

73. TROPOSHERE is the layer of atmosphere above the troposphere but below space

1. True
2. **False**

74. The logical connection between the peer layers is _____ **connection.**

1. Physical
2. **Direct**
3. Indirect
4. Tangible

75. We can have _____ different groups with a 4 bit block.

1. 4
2. **8**
3. 16
4. 32

76. In analog transmission, the base signal is called the _____.

1. **Carrier Signal**
2. Analog signal
3. Digital signal
4. Modulated signal

77. Traditionally MANs have been implemented using one of the 2 technologies, Circuit Switching and Packet Switching
1. true
 2. **false**
78. In_____a dedicated communication path is established between two stations through the nodes of the network
1. **Circuit switching**
 2. Packet switching
79. If the window size is 63 in go-back N ARQ then what is the range of sequence number?
1. **0 to 63**
 2. 0 to 64
 3. 1 to 63
 4. **1 to 64**
80. WDM stands for_____.
1. Wideband De-Modulation
 2. Worst Data Manipulation
 3. **Wavelength Division Multiplexing**
 4. None of the given
81. At the CRC checker,_____means that the data unit is damaged.
1. A string of 0s
 2. A string of 1s
 3. A string of alternating 1s and 0s
 4. **A nonzero remainder**
82. The level of signal is always positive in NRZ encoding.
1. true
 2. **false**
83. There are_____basic categories of multiplexing.
1. 2
 2. **3**
 3. 7
 4. 5
84. _____are used to exchange session management and control information between connected devices.
1. I-frames
 2. **U-frames**
 3. S-frames
 4. N-frames
85. At the CRC generator,_____added to the data unit before the divisional process.

1. 0s are
 2. 1s are
 3. a polynomial is
 4. a CRC remainder is
86. Checksum is an error-detection technique that can be applied to a message of _____ length.
1. Exactly 4
 2. Exactly 8
 3. Exactly 32
 4. Any
87. The _____ protocol uses both flow and error control.
1. TCP/IP
 2. HDLC
 3. ICMP
 4. Stop-and-Wait
88. S-frames in High-Level Data Link Control (HDLC) are only used to transfer _____ information.
1. User data
 2. Control
 3. Redundant
 4. Original
89. Which of the following statement is correct?
1. Digital signals are less prone to Noise
 2. Digital signals are highly prone to Noise
 3. Analog signals are less prone to Noise
 4. Digital signals are not affected by Noise
90. Quantization is a method of assigning _____ values in a specific range to sampled instances.
1. Negative
 2. Positive
 3. Binary
 4. Integral
91. To control and manage the transfer of data, a protocol called Point to Point Protocol (PPP) is used at the _____ layer.
1. Physical
 2. Network
 3. Presentation
 4. Data link
92. _____ is sometimes called the bit rate.
1. signal rate
 2. modulation rate

- 3. Data rate**
4. pulse rate
93. In circuit switched networks we have low efficiency but minimal . 1. Delay
- 2. Speed**
3. Throughput
4. Errors
94. In _____, both peak amplitude and frequency remains constant as the phase changes.
- 1. PSK**
2. QAM
3. Both PSK & QAM
4. None of the given
95. The inversion of the level at 1 bit is called as _____.
1. NRZ-L
- 2. NRZ-I**
3. RZ
4. None of the given.
96. Which error detection method uses one's complement arithmetic.
1. Cyclic Redundancy Check (CRC)
2. Longitudinal Red Check (LRC)
- 3. Checksum**
4. Simple Parity Check
97. The transmission medium that carries the message is referred to as the _____.
1. send and receive device
- 2. communication channel**
3. protocol
4. gateways
98. What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 200, 400, 600 and 800?
- 1. 100**
- 2. 500**
3. 300
4. **600**
99. _____ is the process of converting digital data to digital signals.
1. CSMA/CD
2. CSMA/CA
- 3. Line coding**

4. Segmentation
100. Session layer is responsible for_____.
1. Reassembly of data
 2. **Maintaining the connection**

Error correction is more _____ than the error detection. 1. Easy

Useless

Informal

Difficult

In order to find the location of two errors in an eight bit stream, we have to see _____ different combinations.

28
16
8
4

If users must take turns using the link, then it is called Spatial Sharing.

true

false

_____ is used in the Stop-and-Wait protocol to overcome the issue of duplication.

Frame sequencing

Acknowledgement numbering

Counter and Timer reset

Frame sequencing and Acknowledgement numbering

Asynchronous transmission is_____.

Slow

Fast

Costly

None effective

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What is the frequency of a sine wave if it completes one cycle in 5 seconds?

0.5 2. 0.6

0.3

0.2

RZ stands for_____.

Return to Z

Retire a Zero **Return ot**

Zero None of the given

In Pulse Code Modulation (PCM), the sampling is dependent on_____.

Time

Amplitude

Frequency

Signal Rate

Terminal, microprocessor, computer, printer or any other device that generates or consumes digital data is known as

DCE

DTE

DCE & DTE

None of the given

dB is negative if a signal is_____.

Attenuated

Distorted

Amplified

Noisy

There are three types of Hubs that exist. Which of the following options correctly describes these three?

Passive, dormant, special

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Active, dormant, passive Passive,
Active, Turbo **Passive, Active,**
Intelligent

Data is transmitted using light through a
_____ cable.

twisted pair

fiber-optic

coaxial

microwave

In frequency shift keying, _____ remain(s)
constant.

Frequency Amplitude

Both Amplitude and Phase

Both Phase and Frequency

Human voice is example of _____ signal.

Digital

Analog

Discrete

None of the given

To make sure that source does not overwhelm destination by
sending data faster than it can be handled and processed is
called?

Addressing & Routing

Exchange Management **Flow**

Control Recovery

_____ contains a repeater.

Active hub
Passive hub

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In Block coding we divide our message into blocks of k bits called as _____.

Code words

Data words

Passwords

Cross words

Digital signals are referred to be _____.

Continuous

Discrete

All of the given
None of the given

DLC in Data Link Layer stands for _____.

Divide Line Communication

Data Line Code

Data Link Control

Demand Link Coordination

Digitizing an Analog Signal is called _____.

Analog-to-Digital Conversion

Digital-to-Digital Conversion

Digital-to-Analog Conversion

None of the given

What is the period of a Sine wave having frequency of 10 Hz?

0.2

0.1

0.5

0.3

_____ relatively measures the strength of two signals.

Signal rateBit
rate Decibel

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Pulse Rate

_____ is correct formula to determine the total number of ports needed for one system connected in a mesh network.

$N+1$

$N-1$

$N(N+1)$

$N \times N$

In Block coding scheme, number of code words is always _____ data words.

Less than **Equal**
to Greater than
Same as

Is there any difference between Data communication and telecommunication network?

Yes

No

_____ mode of serial transmission guarantees fixed rate of data.

Synchronous

Asynchronous

Isochronous

Metachronous

In the encoding where positive to negative transition is one and vice versa is zero.

Manchester Differential

Manchester All of the given

None of given

Which of the following is most affected by noise? PSK

ASK

FSK

QAM

The _____ is the number of signal elements sent per unit time.

Data rate

Signal rate Bit

rate Period

Analog signals are referred to be _____

Continuous

Discrete

Periodic

Aperiodic

Guard bands are used in _____ multiplexing technique, to avoid overlapping of frequency bands assigned to each user.

PDM

CSMA

TDM

FDM

There are _____ types of guided media.

3

4

5

_____ of a signal is the collection of all the component frequencies it contains.

Time domain

Frequency domain

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Frequency spectrum
None of the given

Data Link layer provides services to the _____ layer.

Application
Transport
Network
Session

Sampling means measuring _____ of signal at equal intervals

Amplitudes
Frequencies
Phases

None of the given

Which layer is concerned with the syntax and semantics of info exchange between two systems? Physical

Application
Session
Presentation

_____ signal can take infinite levels of intensity over time.

Digital
Discrete
Analog
Logical

Radio is an example of _____ signal conversion.

Analog to analog

Analog to digital Digital

to Digital Digital to

analog

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Which one is not the function of data link layer?

Line discipline Flow

control Error

control Network

control

The most common type of connector used by coaxial cable is

_____.

BNC

RJ-45 3. RJ-11

4. RJ-57

In ASK correct formula for calculating the bandwidth is as__.

$B = (1+d) S$

$B = (1*d) S$

$B = (d-1) S$

$B = (d-5) S$

A portion of the path that carries TX between a given pair of devices is known as_____.

Direct Point

Bridge **Channel**

Access Point

In_____, a signal is directed straight from antenna to antenna.

Line of sight propagation

Ground propagation Space

propagation Tropospheric

propagation

Collisions in CSMA/CA are avoided through _____.

The Interframe Space The
Contention Window

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Acknowledgements

All of the given

A fiber -optic cable transmit signals in the form of

_____.

Light Sound

Wave

None of the given

_____ category of coaxial cable is used for Cable TV.

RG-58

RG-59

RG-11

RG-47

Asynchronous TDM is efficient only when the size of the time slot is kept relatively _____.

Large

Small

Medium

Zero

Following characteristics fully describe

_____,

Amplitude

Period/Frequency

Phase

Sine Waves

YMODEM has _____ 256 3. 128

1024 4. 512

Byte of data unit.

Congestion Control is a feature of layer(s).

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Data Link
Network
Transport

All mentioned

The message 1110 sent by a source is received by a destination as 1011. This is _____ type of error.

Burst

Single-Bit
Uni-Bit
Hamming

Switch is a network device which operates on the _____ layer of the TCP/IP protocol suite.

Application

Transport

Presentation

Data link

AM signal requires _____ the BW of original signal

Twice

Half
Same

Three Times

At the Application layer, object/information is in the form of _____.

Message

Packet

Segment
Frame

What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 300, 600, 900 and 1200?

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100

900

Like Data link layer, _____ is also responsible for Flow control.

Transport Layer
Session Layer

_____ is not a function of Data Link Control. Framing

Modulation

Error Control
Flow Control

VCI address is _____ address in virtual circuit approach.

Private
Variable
Local

Global

STP is more expensive than UTP and is less susceptible to noise.

True
False

WDM stands for _____.

Wave Division Multiplexing Wrong

Division Multiplexing Well Division

Multiplexing West Division

Multiplexing

DSL stands for_____.

Digital Subscriber Line

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_____works with topologies in which one device is designated as a primary station and the other devices are secondary stations.

Polling

MAC stand for?

MAC address is of_____.

48 bits.

Fast Ethernet has a data rate of_____Mbps.

100 Mbps